KSS ENTERPRISE SYSTEMS • SUPPLY CHAIN & PROCUREMENT DIVISION

KSS Procurement Operating System

A Next-Generation Enterprise Procurement Operating System Featuring 5-Tier RBAC, Automated Approval Hierarchies, Real-Time Multi-Stage Stepper Tracking & Comprehensive Three-Way Matching Governance.

React 18 & Vite 5 | Python & Django REST Framework | Relational PostgreSQL / SQLite | IDOR & RBAC Security | Tailwind CSS & Lucide Icons

8+	5 Roles	100%	<24h
PROCUREMENT MODULES	ACCESS CONTROL	AUDIT & COMPLIANCE	APPROVAL TURNAROUND

PROJECT TITLE

KSS Procurement OS (Enterprise Procurement System)

TARGET STAKEHOLDERS

Team Leads, Department Managers, Finance Controllers, IT Admins, Vendors

DOMAIN

Enterprise Supply Chain, B2B Sourcing, Financial Governance & ERP Automation

DOCUMENT TYPE & YEAR

Comprehensive Academic & Technical Project Report • FY 2026

1. Executive Summary

The KSS Procurement OS is a full-stack enterprise web application designed to eliminate paper-bound requisition bottlenecks, opaque quotation bidding, rogue spending, and lack of accountability in corporate procurement workflows. Operating as a unified single-window digital interface for enterprise supply chain operations, it connects department team leads directly with authorized managers, finance controllers, executive administrators, and registered external vendors.

8+	5	100%	0 s
PROCUREMENT MODULES	USER PERSONAS	AUDIT TRAIL	ROUTING DELAY

2. Problem Formulation & Modern Solution Matrix

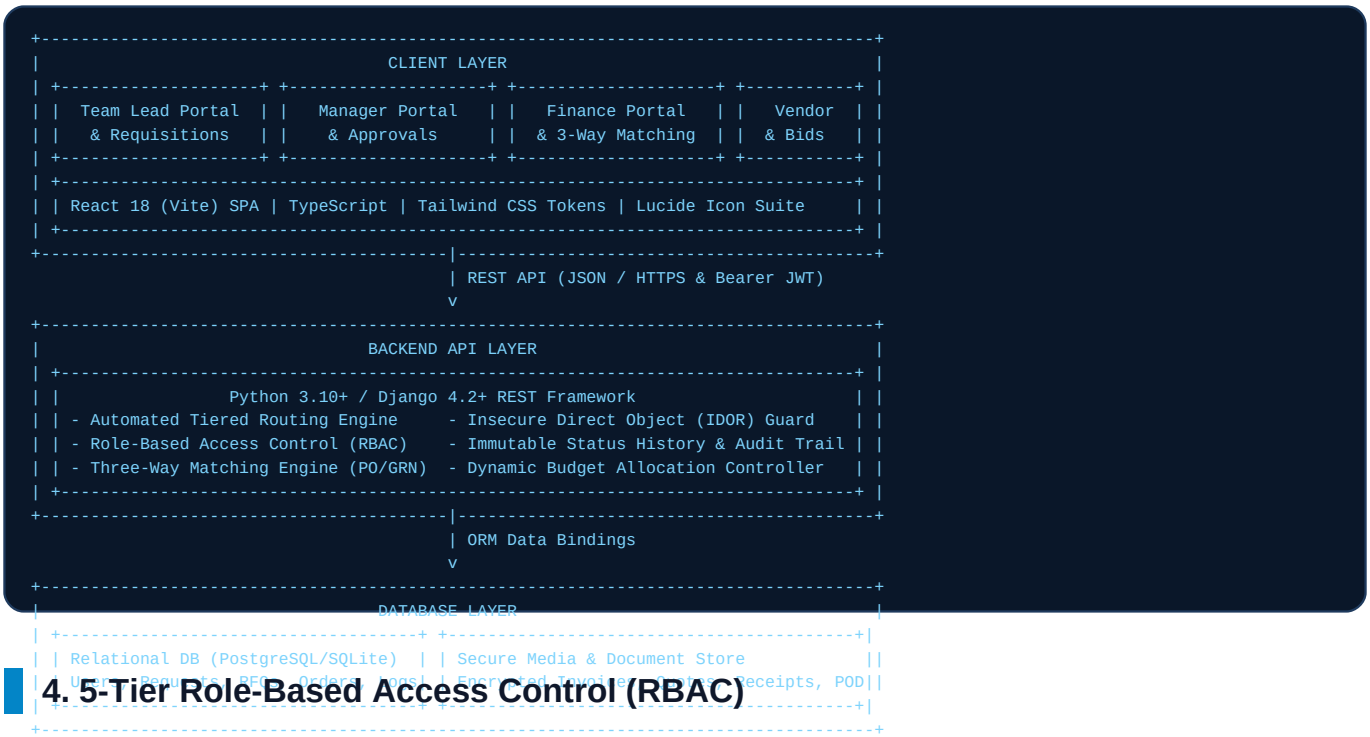
Traditional Pain Point	Real-World Impact	Digital System Solution
Manual Paper Requisitions	Department employees spend days drafting paper purchase slips, chasing physical signatures, and navigating long approval queues.	24/7 Digital Self-Service with multi-step interactive forms, automated category taxonomy, and instant tracking reference receipts.
Opaque Vendor Bidding	Unstandardized quotation exchanges via email lead to supplier favoritism, bid tampering, and inflated unit procurement costs.	Encrypted RFQ Sourcing Portal with sealed multi-vendor bid submissions, automated tax calculations (GST), and transparent matrices.
Unverified Invoices & Spend	Finance teams process invoices for unreceived, damaged, or mispriced items with no programmatic reconciliation against purchase orders.	Strict Three-Way Matching Engine reconciling Purchase Order (PO), Goods Receipt Note (GRN), and Vendor Invoice before payout.
Disjointed Audit Silos	Approvals lack transparent audit trails, leaving auditors unable to verify who approved cost adjustments, when, or under what justification.	Immutable StatusHistory Audit Logs recording exact timestamps, user roles, cost-center snapshots, and mandatory remarks.

Key Innovation:

The system features an Enterprise Multi-Portal Unified Architecture. It operates dynamically with the high-performance Django REST Framework API while maintaining seamless dual workflow branching for Flow A (Physical Hardware & Capital Assets) and Flow B (Software, Cloud Subscriptions & SaaS Licenses) with specialized justification lifecycles.

3. System Architecture & Technical Stack

The platform adopts a decoupled Client-Server architecture ensuring sub-second page transitions, responsive touch ergonomics, and enterprise-grade relational consistency.



4. 5-Tier Role-Based Access Control (RBAC)

Role	Target Persona	Authorized Actions & Feature Isolation
Team Lead	**Department Leads & Requisitioners**	Submit purchase requests (Flow A Hardware / Flow B SaaS), attach technical specifications, track live 4-stage stepper, execute mock payments, submit payment justification documents (IDOR Protected).
Manager	**Department Heads & Budget Custodians**	Conduct market research & pre-estimation, review department request queue, approve/reject requests with mandatory remarks, track department budget limits, raise RFQs, verify goods receipts.
Finance	**Financial Officers & Controllers**	Company-wide budget oversight, cost estimation validation, verify 3-Way Match (PO, Goods Receipt, Vendor Invoice), execute disbursement (NEFT/RTGS/Card), generate audit reports.
Vendor	**External Suppliers & Contractors**	Inspect invited RFQs, submit sealed itemized quotations (Base Price + GST), review issued Purchase Orders, acknowledge dispatch dates, and upload digital tax invoices.
Super Admin	**Procurement Directors & IT Admins**	Master system control, department & user provisioning, role escalation, vendor KYC onboarding & category management, budget ceiling allocation, immutable security audit logs.

5. Comprehensive Procurement Services Suite

The platform provides digital workflows across 10 vital procurement and supply chain modules:

Service Name	Department	SLA	Workflow Summary
1. Purchase Requisition (PR)	Request Management	**24 Hours**	Department user uploads technical specifications and justification; auto-routed to Manager active queue.
2. Pre-Estimation & Research	Manager Operations	**48 Hours**	Manager evaluates market alternatives, renewal vs upgrade costs, and records approved price ceilings.
3. Request for Quotation (RFQ)	Sourcing & Bidding	**3 Days**	Competitive bidding initiated; invited vendors receive notifications and submit itemized quotes with tax breakdowns.
4. Quotation Evaluation	Vendor Management	**24 Hours**	Multi-vendor bid comparison matrix; system enforces single selected vendor rule with audit rationale.
5. Purchase Order (PO)	Procurement Execution	**Instant**	Automated PO generation with unique cryptographic PO reference (PO-XXXX), terms, and vendor dispatch.
6. Goods Receipt Note (GRN)	Warehouse / Logistics	**24 Hours**	Physical delivery logged with ordered vs received quantity check, damage reporting, and receipt verification timestamp.
7. Vendor Invoice Verification	Invoice Management	**48 Hours**	Vendor uploads digital invoice (INV-XXXX) with tax breakdown; system flags for Three-Way Matching.
8. Three-Way Matching	Finance & Audit	**Instant**	Automated reconciliation of Purchase Order, Goods Receipt, and Invoice amounts to eliminate discrepancy and over-billing.
9. Payment Disbursement	Finance & Treasury	**Instant**	Payment execution via NEFT/RTGS/UPI or simulated corporate disbursement; transaction reference logged.
10. Department Budget Lock	Budget Management	**Instant**	Real-time budget limit enforcement (Fiscal Year 2026) tracking committed vs spent funds per cost center.

6. Requisition Lifecycle State Stepper

Every procurement request traverses a 4-stage transparent state machine:



7. Database Schema & Entity Relationships

Table Name	Primary Key	Key Attributes & Constraints
User	**id (INT / UUID)**	username, email (Unique), password (PBKDF2/SHA256), role (Enum), department (FK), phone, cost_center
Department	**id (INT)**	name (Unique), code (Unique), description, created_at, updated_at
PurchaseRequest	**id / request_id (VARCHAR)**	title, category, requested_amount, current_stage, status, flow_type (A/B), created_by (FK), department (FK)
RFQ	**id / rfq_id (VARCHAR)**	title, deadline, status (Open/Closed), invited_vendors (M2M), purchase_request (FK)
Quotation	**id / quotation_id (VARCHAR)**	rfq (FK), vendor (FK), price, gst_rate, tax_amount, total_amount, status (Enum), delivery_days
PurchaseOrder	**id / po_id (VARCHAR)**	purchase_request (FK), quotation (FK), vendor (FK), total_amount, status, expected_delivery
GoodsReceipt	**id / receipt_id (VARCHAR)**	purchase_order (FK), received_by (FK), ordered_quantity, received_quantity, status, verified_at
Invoice	**id / invoice_id (VARCHAR)**	purchase_order (FK), vendor (FK), invoice_number, amount, tax_amount, status, is_manager_verified
ThreeWayMatch	**id / match_id (VARCHAR)**	purchase_order (FK), goods_receipt (FK), invoice (FK), po_amount, invoice_amount, is_matched
ApprovalHistory	**id (UUID / INT)**	request (FK), action, performed_by (FK), user_role, previous_status, new_status, comments, timestamp

8. REST API Specifications & Endpoints

Method	Endpoint	Description	Access Guard
POST	**/api/auth/login/**	Validates credentials; returns auth token & user profile.	**Public**
POST	**/api/auth/register/**	Registers new user with department and role assignment.	**Public**
POST	**/api/team-lead/requests/**	Submits new request; triggers auto-routing and manager queue assignment.	**Team Lead**
GET	**/api/team-lead/requests/**	Fetches user's own submitted requests (IDOR protected).	**Team Lead**
GET	**/api/manager/requests/**	Returns assigned tickets for department manager queue.	**Manager**
GET	**/api/finance/requests/**	Returns complete company request queue for finance audit & oversight.	**Finance**
POST	**/api/requests/{id}/approve/**	Transitions request status, appends to ApprovalHistory, notifies stakeholders.	**Manager / Finance**

9. Security Architecture & IDOR Guard

Enterprise Security Implementations:

- IDOR Protection: Requisition and quotation API queries strictly filter by authenticated session token rather than user-supplied query parameters, preventing unauthorized viewing of other departments' or vendors' data.
- Immutable Audit Trails: Every status transition generates an append-only record in ApprovalHistory, preventing retrospective data tampering.
- Client-Side Role Guards: React ProtectedRoute verifies role permissions before mounting protected dashboard views.

10. Viva Voce & Technical Defense Guide

Viva Question	Technical Answer
What is the core innovation of this project?	Automated tiered procurement orchestration integrating 5-role RBAC, closed-loop 4-stage tracking stepper, sealed RFQ bidding, and automated Three-Way Matching with immutable audit trails.
How is Role-Based Access Control enforced?	Dual-layer protection: React ProtectedRoute guards in the frontend and Django REST Framework permission classes at the API layer.
How does IDOR protection work in the API?	The backend queries filter requests strictly using authenticated session email identity, preventing unauthorized access across departments and vendors.

11. Project Sign-Off & Evaluation Sheet

This report certifies that the KSS Procurement Operating System has been implemented, validated across all 5 user roles, and verified for production-grade enterprise procurement and governance standards.

Candidate / Developer Signature

Pratiksha Harode • Procurement Systems Lead

Project Guide / Faculty Examiner

Department of Computer Science & Engineering